

Explainable AI for Early ICU Deterioration Prediction

Anuja Thuraiyur Jayakumar (25203657)
Ruthvik Gowda Bageri Manjunath (25205410)

Research Question

Can machine learning accurately predict ICU deterioration — while explaining *why*?

Introduction & Motivation

ICU patients can deteriorate rapidly — early prediction enables timely intervention and better outcomes.

- Traditional early-warning scores miss complex, multivariate patterns [2].
- EHRs offer rich, high-frequency patient data — ML can flag high-risk patients *earlier* [2].
- Explainability (XAI) builds the clinical trust needed to act on a prediction [3].

Dataset: MIMIC-IV

Large, public, de-identified critical-care database [1], accessed via Google BigQuery under PhysioNet credentialing.

Predictors: hours 0–6 only — vitals, labs, demographics, care unit.

Cohort: 63,134 first-ICU-stays extracted; **46,982 (74%)** eligible for the 24h task after excluding early deaths and censored stays (discharged before hour 30, no event — ground truth unknown). Deterioration rate: **34.4%**.

Next Steps

- External validation on multi-center eICU-CRD database to test whether performance generalizes across different hospital sites.
- Implement a time-aware Bi-LSTM on raw sequences, to capture fine-grained patient trajectories.
- Clinician-in-the-loop threshold tuning for the alert app, replacing the fixed 0.50 default classification cutoff.
- Extend the 24h horizon sensitivity sweep (36h/48h) with per-horizon hyperparameter tuning.

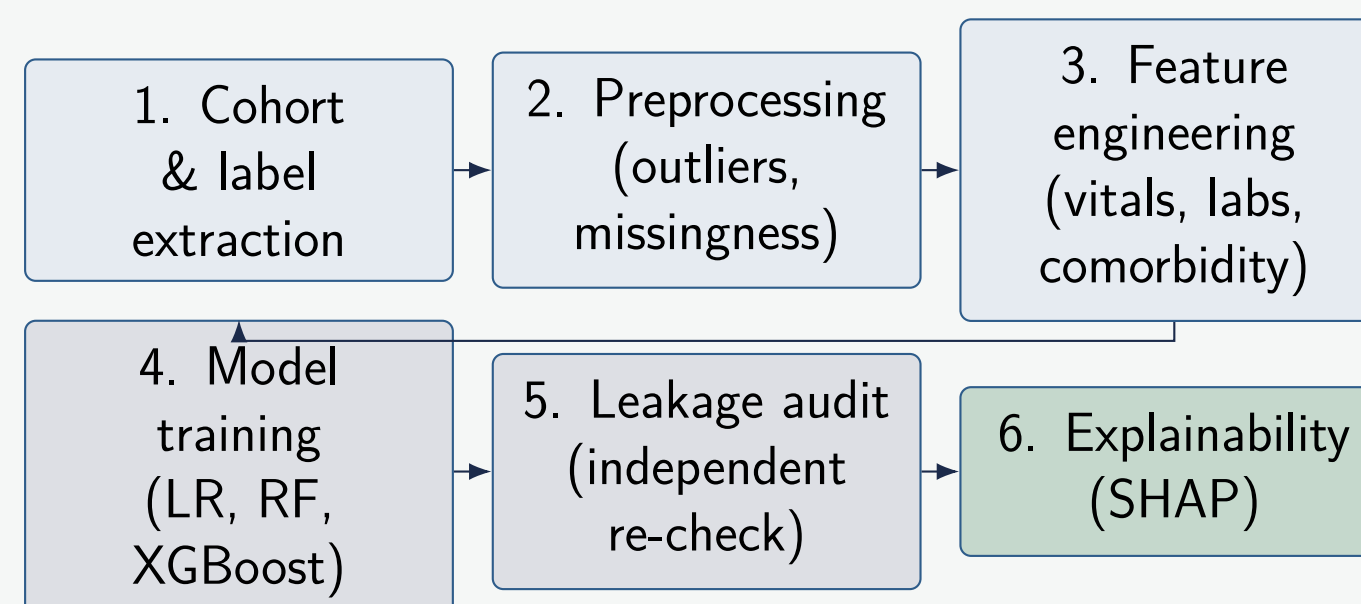
References & Code

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- Churpek, M. M., et al. (2016). Multicenter Comparison of ML Methods for Predicting Clinical Deterioration. *Crit. Care Med.*, 44(2), 368–374.
- Tonekaboni, S., et al. (2019). What Clinicians Want: Contextualizing XAI for Clinical End Use. *PMLR*, 106, 359–380.
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Scan for GitHub Repo

Methodology



Cohort: first ICU stay per patient, ≥ 12 h length of stay.

Label: composite deterioration — mortality OR vasopressor OR ventilation, strictly after hour 6, within 24h.

Leakage prevention: predictors confined to hours 0–6; labels independently re-derived from raw timestamps; zero train/test patient overlap.

ICU admission → **First 6hr: feature observation** → **Subsequent 24hr: deterioration prediction**

Model Performance

80/20 train/test split of the 46,982 eligible stays; hyperparameters tuned via 5-fold CV on the training set only. Held-out test set: $n = 9,397$

Model (24h)	Acc.	Prec.	Rec.	F1	AUC	PR-AUC
Logistic Regression	0.742	0.603	0.731	0.661	0.812	0.692
Random Forest	0.755	0.628	0.701	0.663	0.818	0.702
XGBoost	0.751	0.615	0.736	0.670	0.825	0.716

XGBoost achieved the strongest overall performance, with relatively small differences between the three models.

Results: Horizon Sensitivity

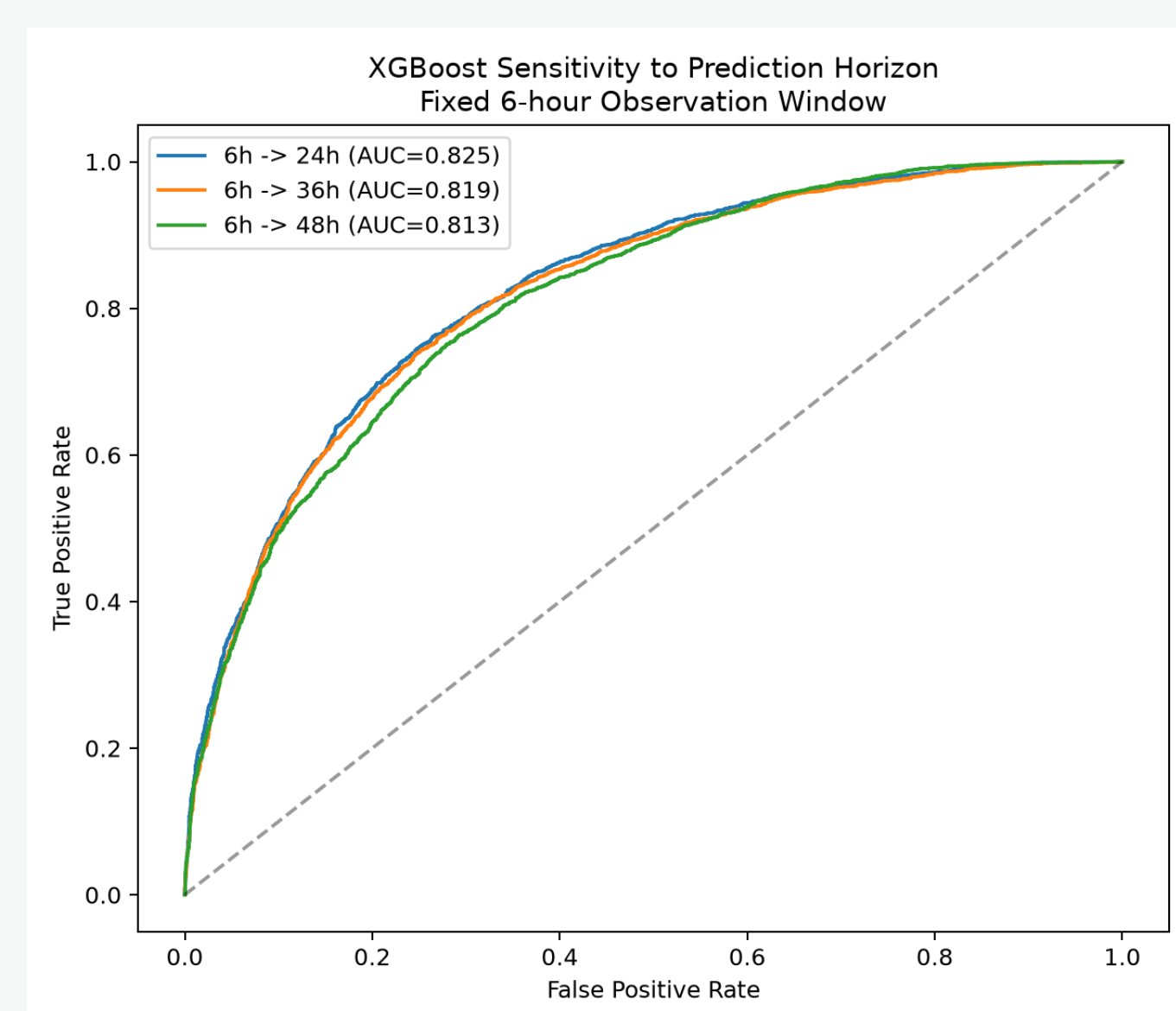


Fig. 3: XGBoost performance across prediction horizons using a fixed 6-h observation window. ROC-AUC declined slightly (0.825 → 0.813), while prevalence increased (34% → 52%). Although PR-AUC increased (0.716 → 0.815), PR-AUC/prevalence lift decreased ($2.08\times \rightarrow 1.57\times$), indicating weaker relative predictive signal at longer horizons.

Calibration

Class-imbalance weighting improves ranking but distorts probabilities (60% predicted → ~45% observed).

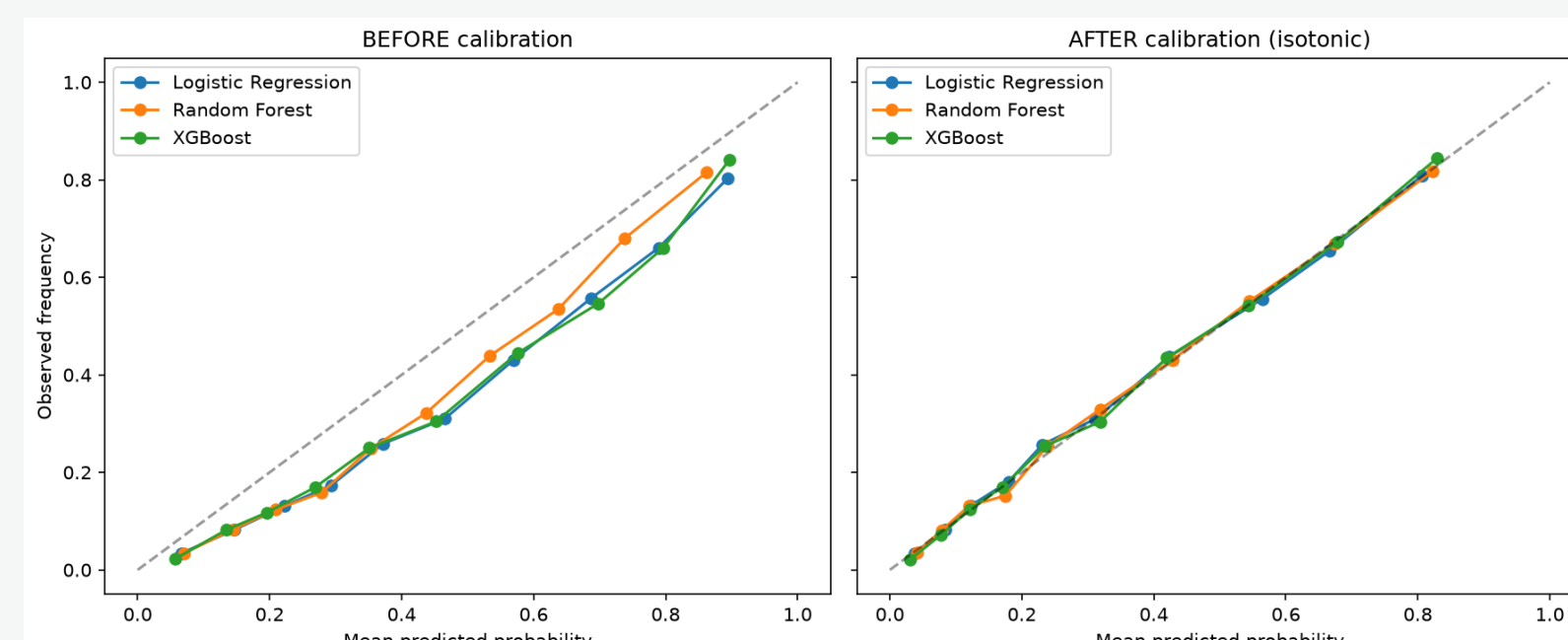


Fig. 4: Isotonic calibration substantially improved agreement between predicted probabilities and observed event frequencies.

Explainability: SHAP

SHAP quantifies each feature's contribution to individual predictions.

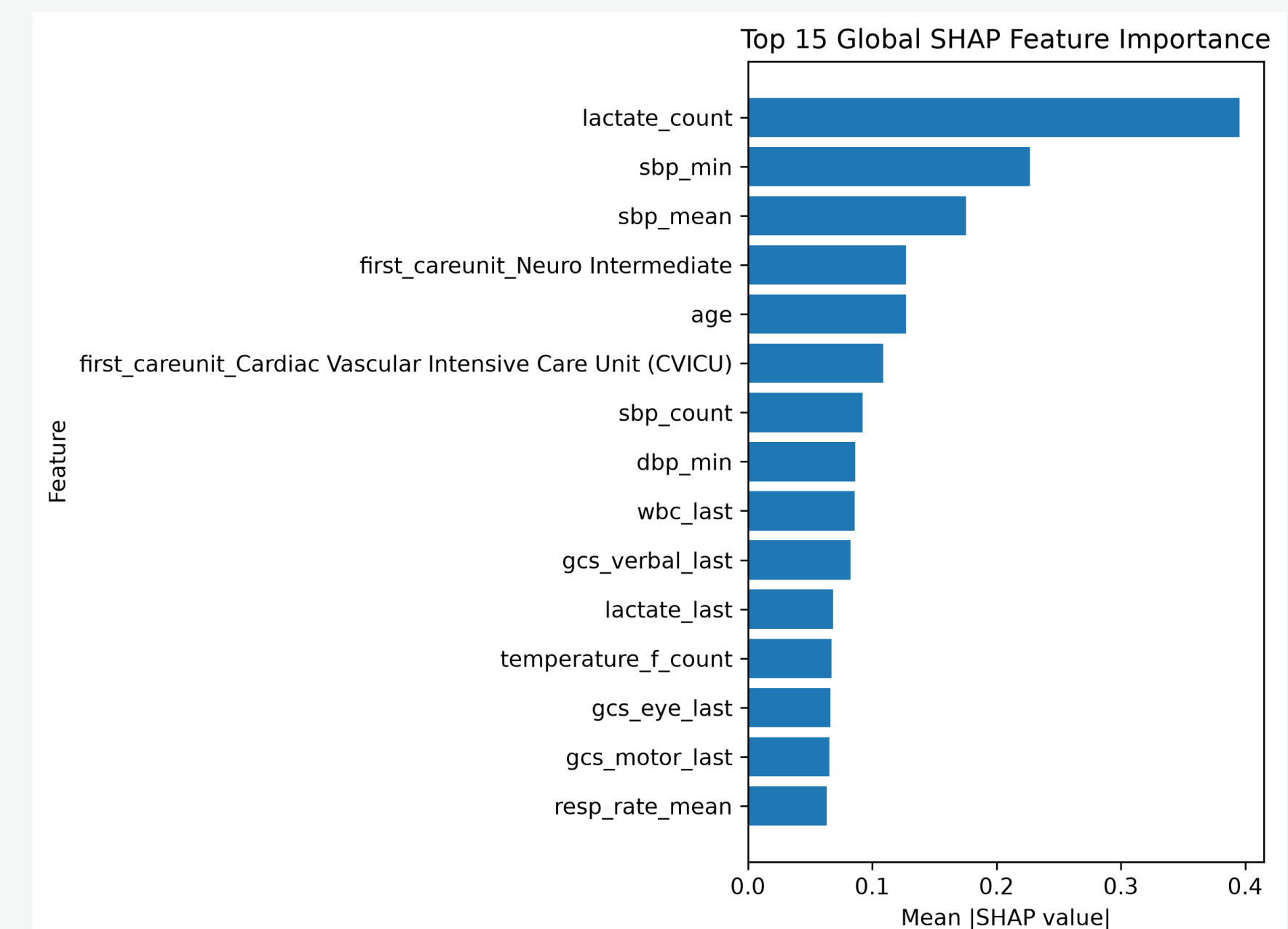


Fig. 5: Global feature importance (XGBoost). Measurement *frequency* (lactate_count) outranks the lactate value itself — sicker patients get tested more often, so testing frequency partly encodes clinician suspicion, not physiology alone.

Patient-Level Explanation

A correctly-caught high-confidence case (raw score +5.21) was driven by low minimum SBP and elevated lactate — textbook physiology. The figure below shows the opposite: the **most-confidently-missed** deterioration (raw score –3.68).

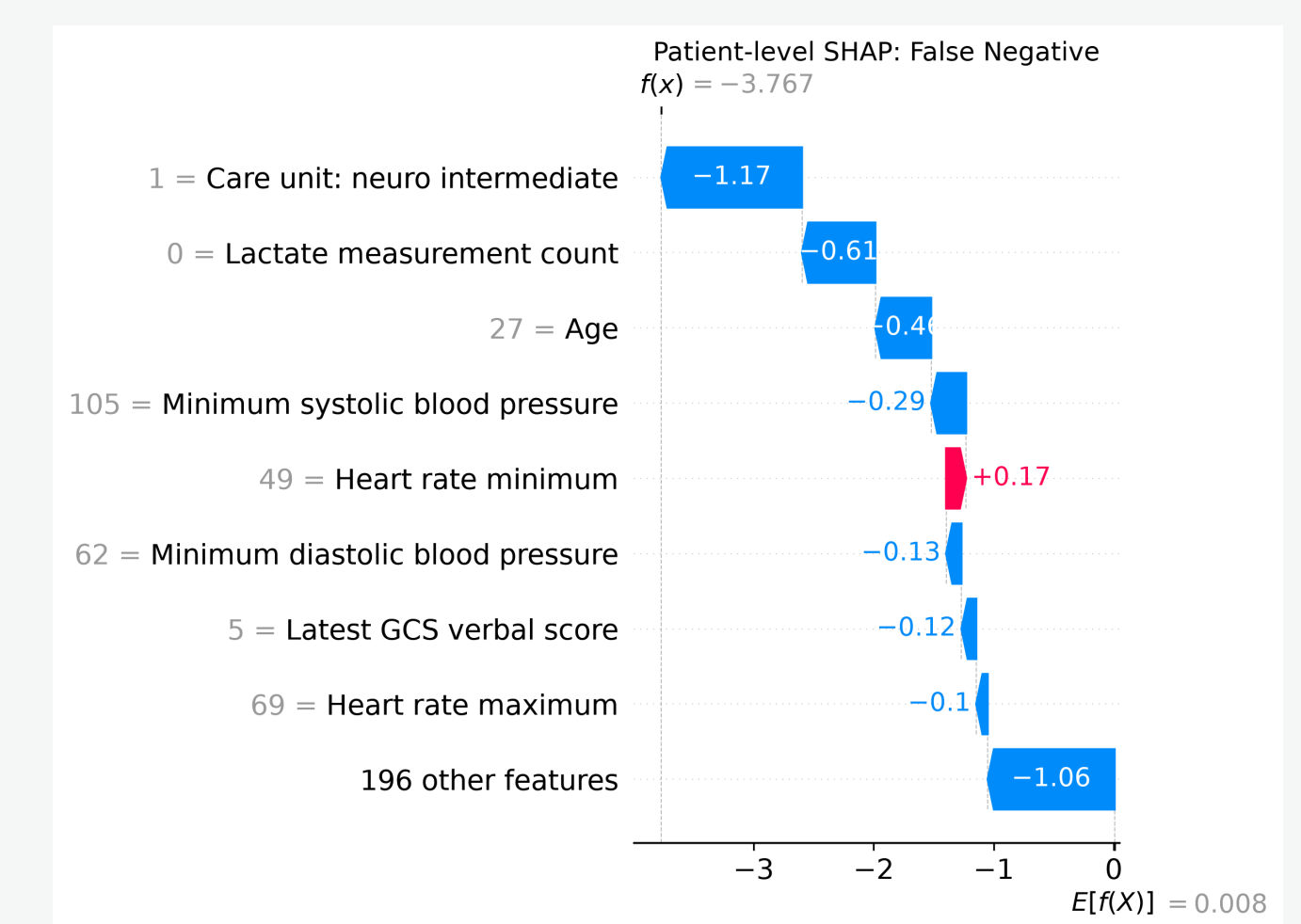


Fig. 6: False-negative case: the model's dominant signal was *admission care unit* (Neuro Intermediate), not this patient's individual vitals or labs — a concrete example of a categorical prior overriding measured physiology for a specific patient.

ICU Deterioration Risk Monitor

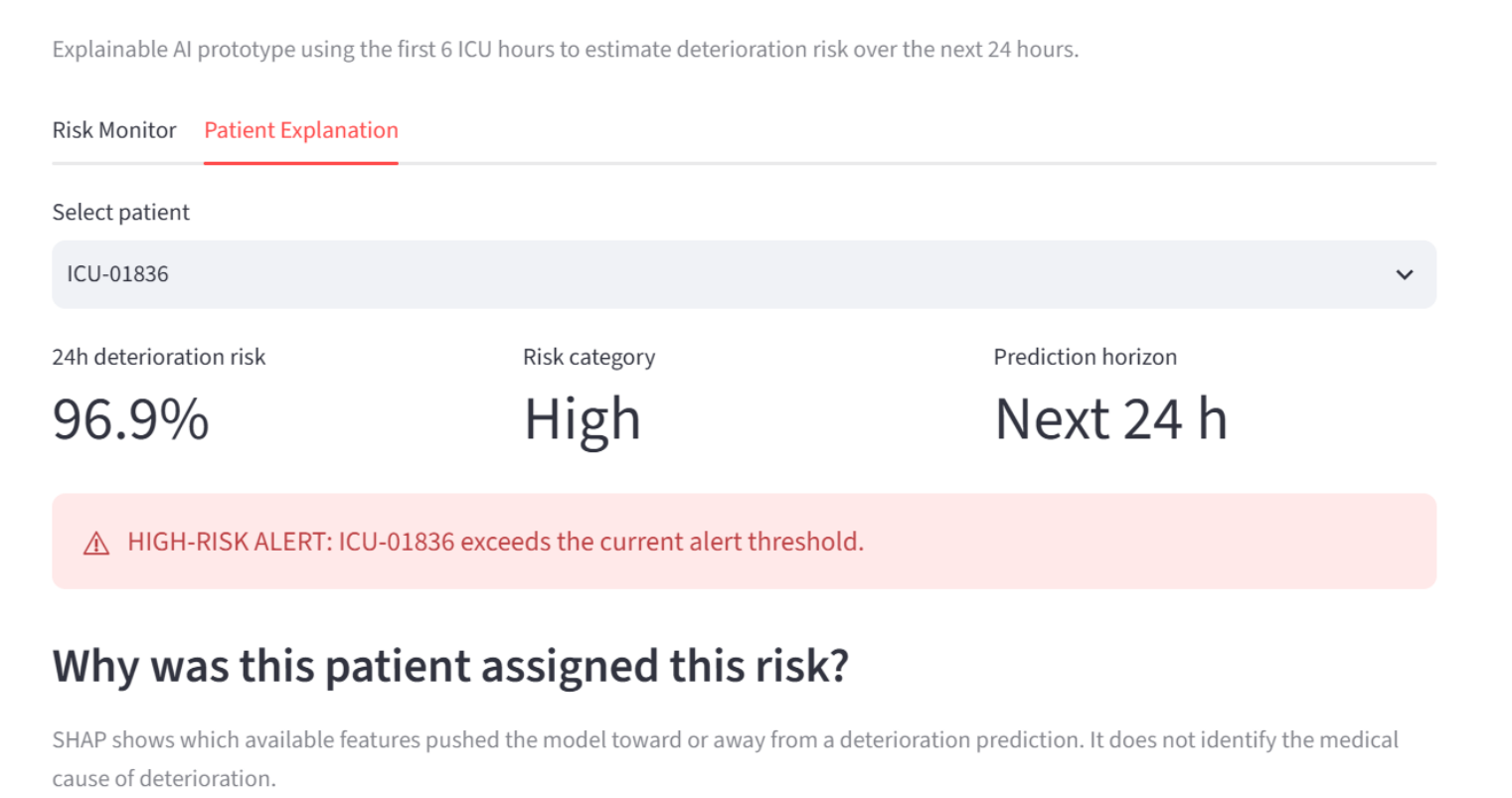


Fig. 7: Interactive prototype presenting calibrated 24-h deterioration risk with patient-level model explanations.

Conclusion

- XGBoost achieved strongest predicted subsequent deterioration with good discrimination, and remained relatively stable at longer horizons, although prevalence-adjusted predictive lift declined.
- SHAP identified clinically plausible physiological patterns, while calibration improved probability reliability.
- Results demonstrate promising and interpretable early-warning performance, but external and prospective validation is required before clinical use.